

THE PARITY CONJECTURE FOR INDEPENDENCE POLYNOMIALS OF GENERALIZED PETERSEN GRAPHS CONTRADICTS ITSELF

DEEP BHATTACHARJEE

ABSTRACT. Pandey conjectured that for $n \geq 2k + 1$ the independence polynomial of the generalized Petersen graph $\text{GP}(n, k)$ has only real roots if and only if k is even. Erlbacher's Demonstrandum project refuted this by exact computation, with $\text{GP}(7, 3)$, $\text{GP}(9, 2)$ and the isomorphism $\text{GP}(7, 2) \cong \text{GP}(7, 3)$. We show that the statement contradicts itself for every $n \equiv 3 \pmod{4}$, since then $\text{GP}(n, 2) \cong \text{GP}(n, (n-1)/2)$ and $(n-1)/2$ is odd, and we give short proofs that each direction fails on its own.

1. INTRODUCTION

For integers $n \geq 3$ and $1 \leq k < n/2$, the generalized Petersen graph $\text{GP}(n, k)$ of Watkins [6] has vertices $u_0, \dots, u_{n-1}$ and $v_0, \dots, v_{n-1}$ and the edges

$$u_i u_{i+1}, \quad u_i v_i, \quad v_i v_{i+k} \quad (0 \leq i < n),$$

with indices read modulo n . The independence polynomial of a graph G is $I(G; x) = \sum_{j \geq 0} a_j x^j$, where a_j is the number of independent sets of G with j vertices; its roots are always real for some families and spread over the plane for others [1]. Real-rootedness of independence polynomials is known for claw-free graphs by the theorem of Chudnovsky and Seymour [2]; generalized Petersen graphs are cubic and contain claws, and their roots were studied numerically by Pandey [4], who proposed the following.

Conjecture 4.1 of [4]. For all integers $n \geq 2k + 1$, the polynomial $I(\text{GP}(n, k); x)$ has only real roots if and only if k is even.

The conjecture assigns a property to the pair (n, k) , whereas real-rootedness depends only on the isomorphism class of $\text{GP}(n, k)$. Different step sizes can give the same graph, and this is enough to refute it.

Theorem 1.1. *The following hold.*

- (a) *If $n \equiv 3 \pmod{4}$ and $n \geq 7$, then $\text{GP}(n, 2) \cong \text{GP}(n, \frac{n-1}{2})$, the step $\frac{n-1}{2}$ is odd, and both pairs $(n, 2)$ and $(n, \frac{n-1}{2})$ satisfy $n \geq 2k + 1$. So Conjecture 4.1 of [4] fails for at least one of these two pairs, for every such n .*
- (b) *$I(\text{GP}(7, 3); x)$ has only real roots, although 3 is odd.*
- (c) *$I(\text{GP}(9, 2); x)$ has exactly five real roots and one pair of non-real roots, although 2 is even.*

Part (a) is a statement about isomorphisms alone and needs no knowledge of the polynomials. Parts (b) and (c) show that neither implication of the conjecture survives separately.

The conjecture was first refuted by the Demonstrandum project of Erlbacher [3], which computed the polynomials of $\text{GP}(7, 3)$ and $\text{GP}(9, 2)$ exactly, counted their real roots with

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Corresponding author: d.bhattacharjee@erl-forschung.de; ORCID [0000-0003-0466-750X](https://orcid.org/0000-0003-0466-750X).

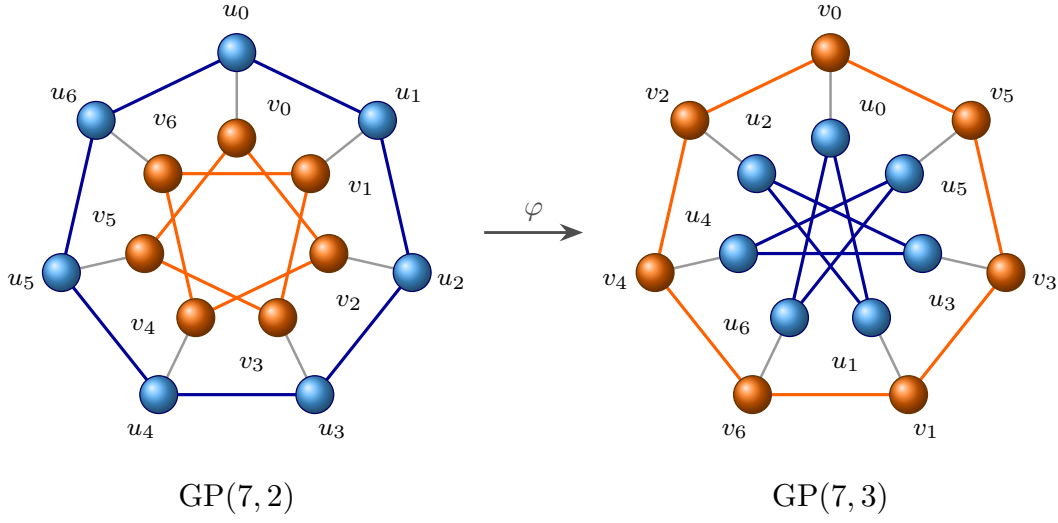


FIGURE 1. The isomorphism φ of Lemma 2.1 for $n = 7$, $k = 2$, $l = 3$. Each vertex of $\text{GP}(7, 3)$ carries the name of its preimage. The outer cycle of $\text{GP}(7, 2)$ (blue) becomes the inner cycle of $\text{GP}(7, 3)$, and the inner cycle (orange) becomes the outer one.

Sturm sequences, and noted the isomorphism $\text{GP}(7, 2) \cong \text{GP}(7, 3)$; it also noted that the triangular prism $\text{GP}(3, 1)$, whose polynomial is $1 + 6x + 6x^2$, is real-rooted. Parts (b) and (c) are therefore theirs, and we include them with proofs that need only sign changes and Vieta's formulas. Part (a) turns their isomorphism into an infinite family.

2. ISOMORPHIC GENERALIZED PETERSEN GRAPHS

Lemma 2.1. *Let $1 \leq k, l < n/2$ with $kl \equiv \pm 1 \pmod{n}$. Then the map $\varphi(u_i) = v_{li}$, $\varphi(v_i) = u_{li}$ is an isomorphism from $\text{GP}(n, k)$ onto $\text{GP}(n, l)$.*

Proof. Since $kl \equiv \pm 1$, the number l is invertible modulo n , so $i \mapsto li$ permutes the indices and φ is a bijection. It maps the outer edge $u_i u_{i+1}$ to $v_{li} v_{li+l}$, an inner edge of $\text{GP}(n, l)$; the spoke $u_i v_i$ to the spoke $v_{li} u_{li}$; and the inner edge $v_i v_{i+k}$ to $u_{li} u_{li+l} = u_{li} u_{li \pm 1}$, an outer edge. Both graphs have $3n$ edges, so φ is an isomorphism. $\square$

Steimle and Staton [5] proved that these maps, together with $\text{GP}(n, k) = \text{GP}(n, n - k)$, account for all isomorphisms between generalized Petersen graphs; we only need the easy direction above.

Proof of Theorem 1.1(a). Let $n \equiv 3 \pmod{4}$, $n \geq 7$, and $l = (n - 1)/2$. Then $2l = n - 1 \equiv -1 \pmod{n}$, so $\text{GP}(n, 2) \cong \text{GP}(n, l)$ by Lemma 2.1. Since $n - 1 \equiv 2 \pmod{4}$, l is odd, and $n = 2l + 1 \geq 2 \cdot 2 + 1$. Conjecture 4.1 of [4] for $(n, 2)$ says that $I(\text{GP}(n, 2); x)$ is real-rooted, and for (n, l) that the same polynomial is not. $\square$

The same argument applies whenever some even k and odd l below $n/2$ satisfy $kl \equiv \pm 1 \pmod{n}$; for instance $4 \cdot 3 \equiv -1 \pmod{13}$ gives $\text{GP}(13, 4) \cong \text{GP}(13, 3)$.

3. THE TWO POLYNOMIALS

Counting independent sets of each size gives

$$(3.1) \quad I(\text{GP}(7, 2); x) = 1 + 14x + 70x^2 + 154x^3 + 147x^4 + 49x^5,$$

$$(3.2) \quad I(\text{GP}(9, 2); x) = 1 + 18x + 126x^2 + 438x^3 + 801x^4 + 747x^5 + 303x^6 + 27x^7.$$

Some coefficients can be read off at once: a_1 is the number of vertices and $a_2 = \binom{2n}{2} - 3n$, since $\text{GP}(n, k)$ has $3n$ edges. Since $2 \cdot 3 \equiv -1 \pmod{7}$, Lemma 2.1 gives $\text{GP}(7, 3) \cong \text{GP}(7, 2)$, so (3.1) is also the independence polynomial of $\text{GP}(7, 3)$ (Figure 1).

Proof of Theorem 1.1(b). Let P be the polynomial (3.1). Its values at seven points are

x	$-\frac{7}{5}$	$-\frac{13}{10}$	-1	$-\frac{3}{5}$	$-\frac{3}{10}$	$-\frac{1}{5}$	$-\frac{1}{10}$
$P(x)$	$-\frac{8733}{3125}$	$\frac{67513}{10^5}$	1	$-\frac{5}{697}$	$\frac{1363}{10^5}$	$-\frac{5}{39}$	$\frac{16021}{10^5}$

The sign changes five times along the table, so P has a root in each of the five disjoint intervals $(-\frac{7}{5}, -\frac{13}{10})$, $(-1, -\frac{3}{5})$, $(-\frac{3}{5}, -\frac{3}{10})$, $(-\frac{3}{10}, -\frac{1}{5})$ and $(-\frac{1}{5}, -\frac{1}{10})$. As P has degree five, all its roots are real. By the remark above, $P = I(\text{GP}(7, 3); x)$, and 3 is odd. $\square$

Proof of Theorem 1.1(c). Let Q be the polynomial (3.2). Its sign at the ten points below changes across each of the five intervals

$$\begin{aligned} Q(-8.28862) < 0 < Q(-8.28861), & \quad Q(-0.51657) > 0 > Q(-0.51656), \\ Q(-0.31262) < 0 < Q(-0.31261), & \quad Q(-0.22263) > 0 > Q(-0.22262), \\ Q(-0.16871) < 0 < Q(-0.16870), \end{aligned}$$

so Q has real roots $\rho_1, \dots, \rho_5$, one in each interval. Let r and r' be the two remaining roots, counted with multiplicity. Since Q has degree seven with leading coefficient 27, the relations between roots and coefficients give

$$r + r' = -\frac{303}{27} - \sum_{i=1}^5 \rho_i, \quad rr' = -\frac{1}{27 \rho_1 \rho_2 \rho_3 \rho_4 \rho_5}.$$

Both numbers are real. Adding the endpoints of the intervals gives $-9.50915 < \sum \rho_i < -9.50910$, so $-1.713123 < r + r' < -1.713071$ and $(r + r')^2 < 2.9348$. All ρ_i are negative and $|\rho_1 \cdots \rho_5| < 8.28862 \cdot 0.51657 \cdot 0.31262 \cdot 0.22263 \cdot 0.16871 < 0.050278$, so $rr' > 1/(27 \cdot 0.050278) > 0.7366$ and $4rr' > 2.9464$. Hence

$$(r - r')^2 = (r + r')^2 - 4rr' < 0,$$

so $r - r'$ is not real, and r and r' are complex conjugates that are not real. Thus $Q = I(\text{GP}(9, 2); x)$ has exactly five real roots, although 2 is even. $\square$

Remark 3.1. The non-real pair of $I(\text{GP}(9, 2); x)$ is close to $-0.85655 \pm 0.05554i$. In [4] the polynomials were computed with a transfer matrix whose entries carry the weight $x^{|B|}$, where $|B|$ is the number of occupied positions in a state that also records earlier inner vertices; with this weighting a vertex is counted once for every state in which it appears. The report there of 50 roots for $\text{GP}(25, 2)$ reflects this: the spokes $u_i v_i$ cover every vertex and an independent set meets each spoke at most once, so $I(\text{GP}(25, 2); x)$ has degree at most 25. The polynomials behind the numerical evidence in [4] are therefore not independence polynomials.

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FORMERLY, ELECTRO-GRAVITATIONAL SPACE PROPULSION LABORATORY (EGSPL),
BHUBANESWAR, ODISHA 751030, INDIA

Email address: d.bhattacharjee@erl-forschung.de

Email address: itsdeep@live.com

URL: <https://orcid.org/0000-0003-0466-750X>